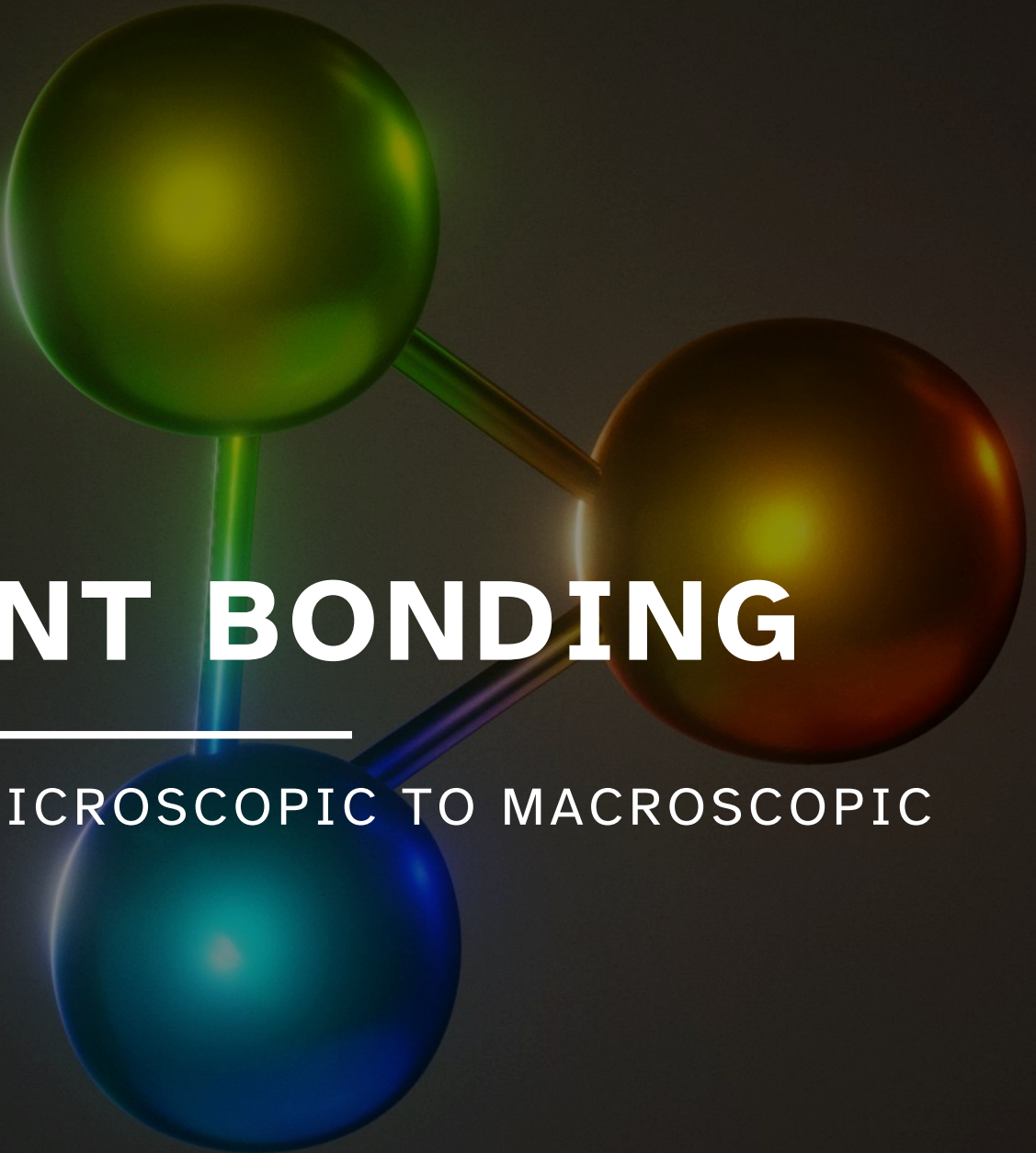




10B COVALENT BONDING

UNIT 04: FROM MICROSCOPIC TO MACROSCOPIC

CHEMISTRY 1310

LEARNING OBJECTIVES

CHAPTER 10

1. Predict the number of bonds an atom is likely to form in a molecule.
2. Draw Lewis structures for molecules that follow the octet rule.
3. Define and apply electronegativity and its periodic trends.
4. Draw Lewis structures for molecules for which multiple valid Lewis structures exist (called resonance).
5. Draw Lewis structures for molecules that do not obey the octet rule by application of the typical exceptions exhibited by H, Be, B, and expanded octets.
6. Determine bond polarity using electronegativity values.
7. Describe and apply the relation between bond length and bond strength.
8. Apply bond enthalpies to calculate enthalpies of reaction.

Exceptions to the Octet Rule

10.4

Limitations/Drawbacks to Lewis structure theory and “bandage solutions”

- 1) Electrons are treated as even pairs of electrons: 2, 4, 6, 8
 - What about molecules with an *odd number of electrons*?
 - What about radicals (molecules with *unpaired electrons*)?
 - We will not deal with these cases.
- 2) Resonance:
- 3) Incomplete octet:
- 4) Expanded octet:

Let's work through these more “exotic” molecules' Lewis structures!

Resonance and Electronegativity

10.3 (we are not covering formal charges)

In some cases, a molecule cannot be represented accurately by a *single* Lewis structure but rather by *multiple valid* Lewis structures.

- These molecules/ions are said to exhibit _____.
- The individual Lewis structures are called _____.
- These structures differ *only* in the placement of the double bonds, triple bonds, and/or lone pairs.

Resonance and Electronegativity

10.3 (we are not covering formal charges)

Misconceptions: The individual resonance structures do not exist. The double-headed arrow does not mean that the molecule alternates between the resonance structures.

The *real/true structure* of the molecule is called the _____.

- It is constructed by taking a *geometric average* over all resonance structures.
- The pairs of electrons involved in the resonance structures are said to be “spread out” among the different positions:

EXAMPLE PROBLEM

Learning Objective(s): 10.2, 10.4

Draw three *different* Lewis structures for CO₂.

Exceptions to the Octet Rule

10.4

Incomplete octets

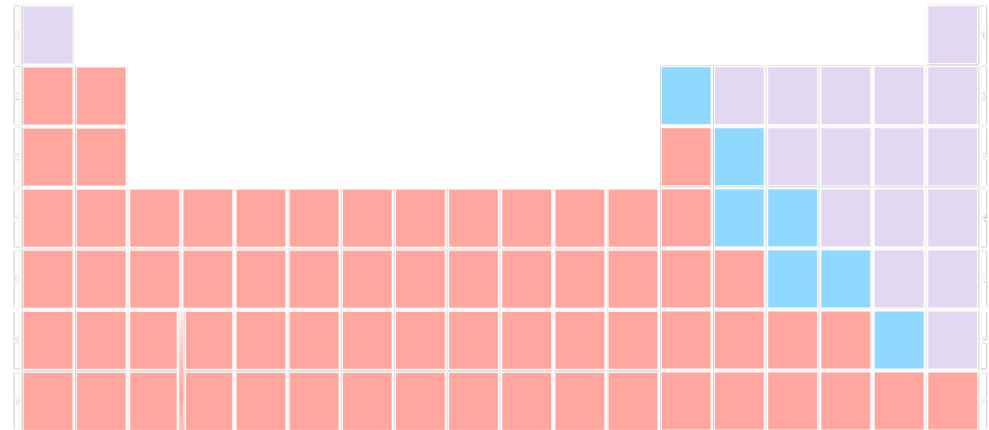
- **H** and **He**: only needs 2 e^- (recall duet rule) H_2O
- Beryllium, **Be**: BeF_2
- Boron, **B**: BH_3

Exceptions to the Octet Rule

10.4

Expanded octets

- The **central atom** can expand or exceed the octet rule **IF it must**.
 - The central atom must be from **periods/rows**:
 - Never expand the octets for outer atoms.



Resonance and Electronegativity

10.3 (we are not covering formal charges)

Electronegativity (EN)

- The ability of an atom to attract *shared* electrons to itself in a bond.

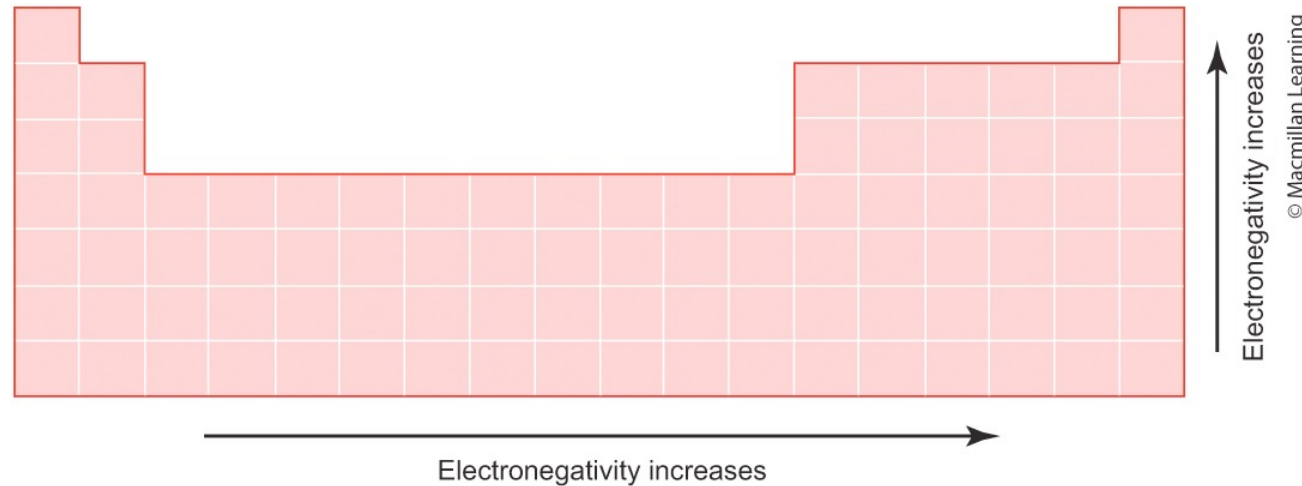


FIGURE 11.15 General electronegativity trend

- Different numerical scales exist, but Pauling scale is most common.

Polar Bonds and the Bonding Continuum

10.5

The difference in electronegativity (ΔEN) between two atoms in a bond can be used to classify bonds by ionic character on a continuum.

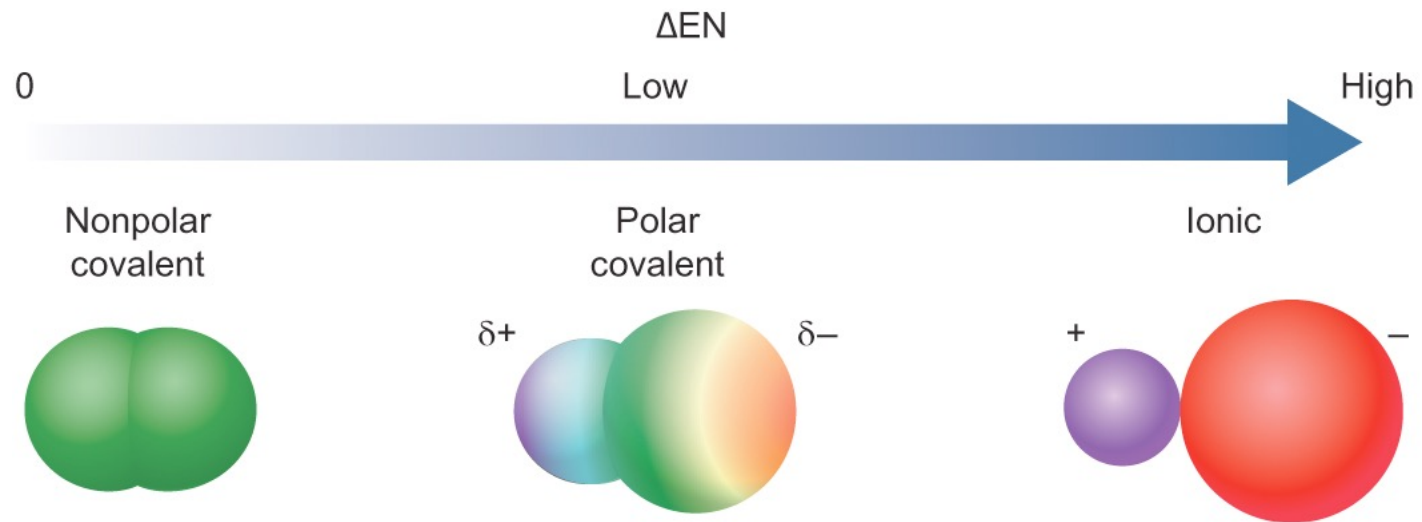


FIGURE 10.17 The bonding continuum

Polar Bonds and the Bonding Continuum

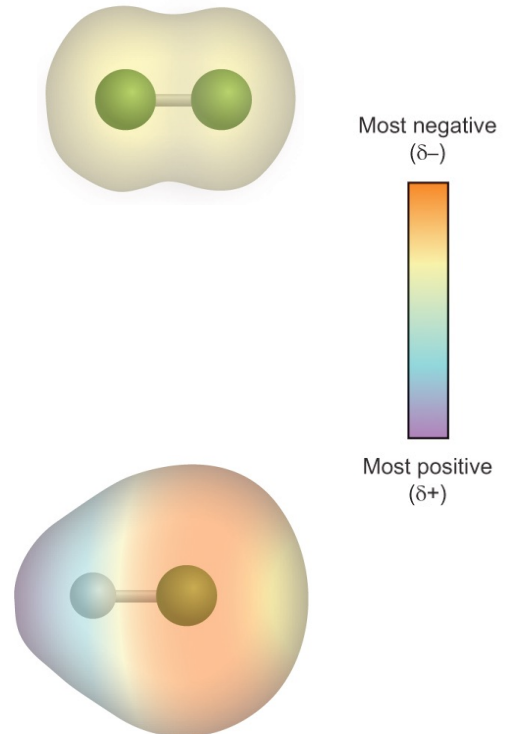
10.5

Nonpolar covalent bond

- Bond between two atoms of _____ electronegativities (EN).
- The electrons are shared _____.

Polar covalent bond

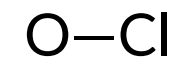
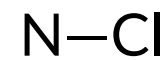
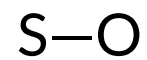
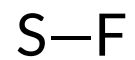
- Bond between two atoms of _____ electronegativities (EN).
- The electrons are shared _____.
- There is a **partial positive charge ($\delta+$)** on the *less* EN atom.
- There is a **partial negative charge ($\delta-$)** on the *more* EN atom.
- The $\delta+$ and $\delta-$ constitute a bond dipole.



IN-CLASS QUESTION 1

Learning Objective(s): 10.6

Rank the bonds in order of *increasing* ionic character.



1 1A								18 8A
H 2.1	2 2A		13 3A	14 4A	15 5A	16 6A	17 7A	He
Li 1.0	Be 1.5		B 2.0	C 2.5	N 3.0	O 3.5	F 4.0	Ne
Na 0.9	Mg 1.2		Al 1.5	Si 1.8	P 2.1	S 2.5	Cl 3.0	Ar
K 0.8	Ca 1.0		Ga 1.6	Ge 1.8	As 2.0	Se 2.4	Br 2.8	Kr
Rb 0.8	Sr 1.0		In 1.7	Sn 1.8	Sb 1.9	Te 2.1	I 2.5	Xe
Cs 0.7	Ba 0.9		Tl 1.8	Pb 1.9	Bi 1.9	Po 2.0	At 2.2	Rn
Fr 0.7	Ra 0.9							

Electronegativity increases

Electronegativity increases

Where did we start and where are we going?

Chemistry in Context

What is our overall goal?



Bond Enthalpy

10.6

Bond enthalpy or bond energy (ΔH_{BE})

- The enthalpy change associated with breaking a specific bond in 1 mol of gaseous molecules.
- Bond breaking is always *endothermic*, so all $\Delta H_{\text{BE}} > 0$ (positive values).
- Stronger bonds are more stable and have larger bond enthalpies (ΔH_{BE}).

Bond Enthalpy

10.6

General trends

- Polar bonds tend to be _____ than nonpolar bonds.
- Multiple bonds (= and $\equiv$) are _____ than single bonds (—).
- Stronger bonds tend to be _____.
- Weaker bonds tend to be _____.

Bond	# of Electrons	Bond Enthalpy (ΔH_{BE})	Bond Length
C—C	2 e^-	348 kJ/mol	154 pm
C=C	4 e^-	614 kJ/mol	134 pm
C $\equiv$ C	6 e^-	839 kJ/mol	120 pm

Bond Enthalpy

10.6

We can use bond enthalpies to contextualize/understand ΔH_{rxn} .

- Bond *breaking* is
- Bond *formation* is

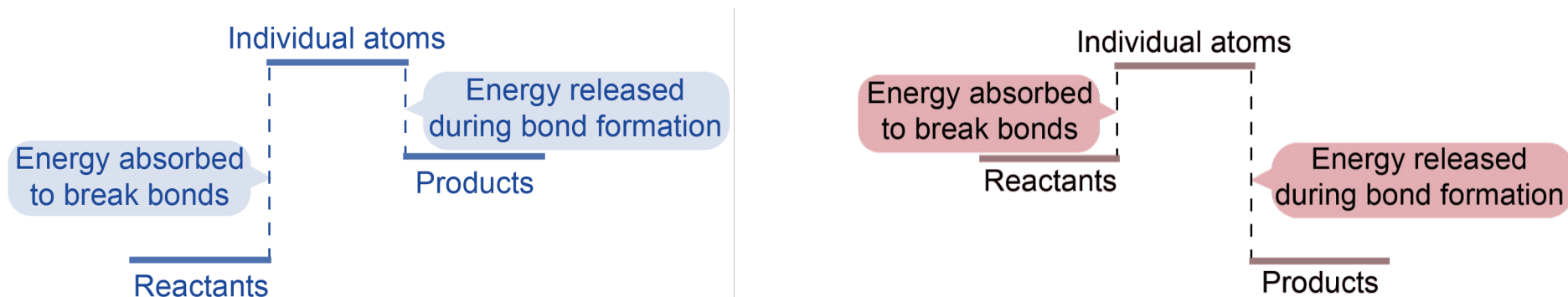


FIGURE 10.19

Bond Enthalpy

10.6

In some cases, we can even use bond enthalpies to *estimate* ΔH_{rxn} .

- Difference between energy put into breaking bonds in reactants and energy released when forming new bonds in the products is *approximately* ΔH_{rxn} .

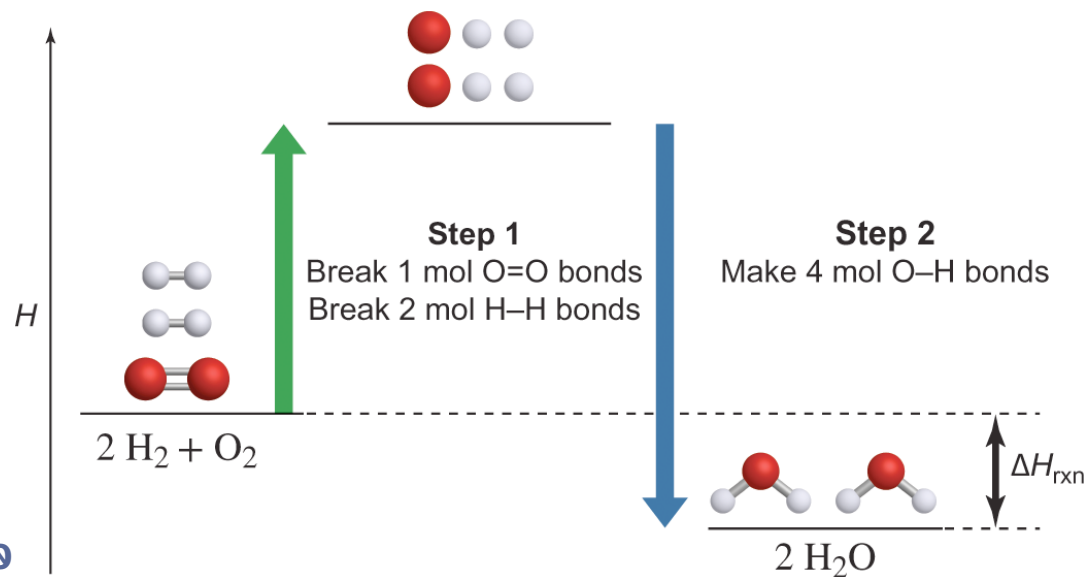
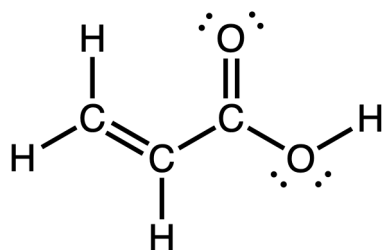


FIGURE 10.20

IN-CLASS QUESTION 2

Learning Objective(s): 10.12

What is the standard change in enthalpy ($\Delta H_{\text{rxn}}^{\circ}$) for the combustion of acrylic acid ($\text{C}_3\text{H}_4\text{O}_2$)?

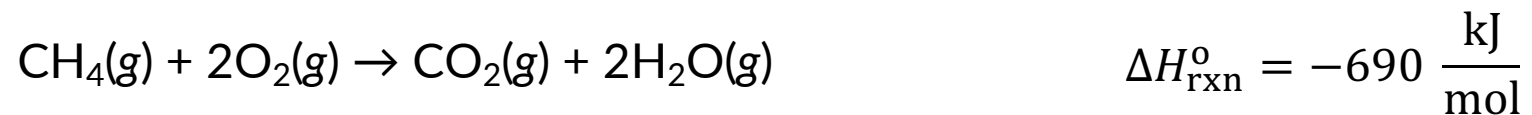


Bond	Bond Enthalpy
H–H	436 kJ/mol
H–C	414 kJ/mol
H–O	464 kJ/mol
C–C	347 kJ/mol
C=C	611 kJ/mol
C–O	360 kJ/mol
C=O	799 kJ/mol
O–O	142 kJ/mol
O=O	498 kJ/mol

EXAMPLE PROBLEM

Learning Objective(s): 10.12

The thermochemical equation for the combustion of methane is given below.



Use this information and the table of bond enthalpies to estimate the bond enthalpy for a C–H bond.

Bond	Bond Enthalpy
C–H	
O=O	496 kJ/mol
C=O	743 kJ/mol
H–O	463 kJ/mol

BONUS QUESTION

Assessed on: Fall 2021, Exam 3

Which elements could be X in the Lewis structure drawn? Select all that apply.

Hint: There are only two correct answers.

☐ C

☐ N

☐ P

☐ Ar

☐ As

